Student Segmentation and Personalized Course Recommendation System for EduPro

Research Paper

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Abstract

This research paper presents a data-driven student segmentation and personalized course recommendation system designed for the EduPro online learning platform. The study analyzes behavioral, demographic, and transactional data from 3,000 learners enrolled across 200 courses, encompassing 11,470 transactions. Using unsupervised machine learning techniques — specifically K-Means clustering validated against Agglomerative Hierarchical Clustering — learners are segmented into four distinct behavioral profiles: Casual Explorers, Power Learners, Budget Browsers, and Focused Achievers. A cluster-aware recommendation engine is then developed using content-based filtering combined with rating-weighted relevance scoring. Evaluation via Silhouette Score and Engagement Lift metrics demonstrates the system's effectiveness in capturing meaningful behavioral differences across learner groups. The Streamlit web application provides an interactive interface for learner profile exploration, cluster visualization, and real-time recommendation delivery.

Keywords: student segmentation, K-Means clustering, personalized recommendation, e-learning, content-based filtering, learner profiling, EduPro

1. Introduction

1.1 Background and Context

The rapid expansion of online education has transformed how learners engage with educational content. Platforms such as EduPro serve diverse learner populations ranging from career-switchers and hobbyists to working professionals seeking specialization. Unlike a traditional classroom, an online platform must accommodate thousands of unique learning styles, pacing preferences, and knowledge goals simultaneously.

Research consistently shows that generic, one-size-fits-all recommendation approaches fail to maximize learner engagement, course completion rates, or long-term platform retention. Learners who receive irrelevant recommendations disengage faster and exhibit lower lifetime value to the platform. Conversely, highly personalized recommendation engines — such as those employed by Netflix and Spotify — have demonstrated significant improvements in user engagement and retention.

1.2 Problem Statement

EduPro currently faces three core challenges:

- One-size-fits-all course recommendations that ignore learner behavior and preferences
- Limited understanding of distinct learner behavior patterns across the user base
- No structured learner segmentation framework to enable targeted marketing or personalized learning journeys

As a result, learners struggle to discover relevant content, and engagement and retention opportunities are lost.

1.3 Objectives

- Perform comprehensive exploratory data analysis on the EduPro dataset
- Engineer meaningful learner-level behavioral and preference features
- Apply unsupervised clustering algorithms to identify distinct learner segments
- Build a personalized course recommendation engine tailored to each segment
- Deploy an interactive Streamlit dashboard for stakeholder access

2. Literature Review

Personalized learning systems have been extensively studied in the educational data mining (EDM) and learning analytics (LA) communities. Early works by Baker & Yacef (2009) established foundational EDM methodologies for understanding student behavior patterns. Collaborative filtering, pioneered by Goldberg et al. (1992), forms the backbone of most commercial recommendation systems. However, pure collaborative filtering suffers from the cold-start problem — new users with no history receive poor recommendations.

Content-based filtering (CBF) addresses this by matching learner preferences to item attributes, making it suitable for sparse interaction data typical of e-learning platforms. Hybrid approaches combining CBF with cluster-based user profiling have shown superior performance in educational contexts (Klasnja-Milicevic et al., 2017). K-Means clustering remains the dominant technique for learner segmentation due to its scalability and interpretability, with silhouette analysis providing reliable cluster quality assessment (Rousseeuw, 1987).

More recent work by Nabizadeh et al. (2020) on adaptive learning paths and by Thai-Nghe et al. on matrix factorization for e-learning recommendation further motivates a hybrid, cluster-aware approach that balances scalability with personalization depth — the approach adopted in this study.

3. Dataset Description

3.1 Data Sources

The EduPro dataset comprises three relational tables:

Sheet	Records	Key Fields
Users	3,000	UserID, Age, Gender, Email
Courses	200	CourseID, CourseCategory, CourseType, CourseLevel, CourseRating
Transactions	11,470	UserID, CourseID, TransactionDate, Amount

3.2 Descriptive Statistics

- Users: Age range 15–60, mean age ~25 years, gender split approximately 50/50
- Courses: 8 categories (AI/ML, Data Science, Web Development, Business, Design, Marketing, Cybersecurity, Finance), 3 levels (Beginner/Intermediate/Advanced), ratings 2.5–5.0
- Transactions: Average 3.8 enrollments per user, amounts $0–$200, spanning 2023–2024

3.3 Data Quality

No missing values were identified across all three tables. Categorical fields were validated for consistency. Transaction dates were converted to datetime format for temporal analysis. All numerical features were normalized prior to clustering using StandardScaler.

4. Methodology

4.1 Feature Engineering

Learner-level features were engineered across three categories:

Feature Category	Features	Purpose
Engagement	Total courses enrolled, Enrollment frequency, Total spending, Avg spending/course	Measures activity level and financial commitment
Preference	Preferred category, Preferred level, Category diversity, Avg rating enrolled	Captures content affinity and breadth

Behavioral	Learning depth index (Advanced/Beginner ratio), Diversity score	Indicates learning maturity and exploration style

4.2 Data Preprocessing

- Categorical encoding: LabelEncoder applied to preferred_category, preferred_level, and gender
- Normalization: StandardScaler applied to all 10 numeric features to ensure equal contribution
- Noise reduction: Sparse enrollment patterns smoothed via aggregation at UserID level

4.3 Clustering Approach

K-Means clustering was selected as the primary algorithm due to its computational efficiency on 3,000 samples and interpretable centroid-based profiles. The Elbow Method was used to assess inertia across k=2 to k=8, and Silhouette Scores were computed for each k to identify the optimal cluster count. k=4 was selected, balancing statistical optimization with business interpretability.

Agglomerative Hierarchical Clustering (Ward linkage) was applied at k=4 as a validation method. The silhouette scores from both methods were compared to confirm cluster stability.

4.4 Recommendation Engine

The recommendation engine combines three signals:

- Content-based filtering: Matches unenrolled courses to the learner's preferred category (+3 pts) and preferred level (+2 pts)
- Course quality: Rating-weighted relevance (CourseRating / 5.0) rewards highly-rated courses
- Cluster-based popularity: Counts how many cluster peers enrolled in each course, normalized to [0,1]

Final score = category_match(3) + level_match(2) + rating_score(0–1) + peer_popularity(0–1). Courses already enrolled by the user are excluded.

5. Results and Analysis

5.1 Cluster Profiles

Segment	Size	Avg Courses	Avg Spend	Top Category	Top Level
Casual Explorers	921 (30.7%)	3.5	$375	AI/ML	Beginner
Power Learners	675 (22.5%)	6.1	$622	AI/ML	Advanced
Budget Browsers	640 (21.3%)	2.3	$156	Cybersecurity	Advanced
Focused Achievers	764 (25.5%)	3.5	$361	AI/ML	Advanced

5.2 Cluster Interpretations

Casual Explorers (30.7%): The largest segment, moderate enrollment (3.5 avg), limited spending, beginner-level preferences. Learners are sampling the platform without deep commitment. Strategy: onboarding nudges, free introductory courses, and gamification elements.

Power Learners (22.5%): Highest-value segment with 6.1 avg courses and $622 avg spending. Gravitate toward advanced content and exhibit high category diversity. Strategy: subscription bundles, advanced certification tracks, and cohort-based programs.

Budget Browsers (21.3%): Low enrollment (2.3 courses) and minimal spending ($156). Prefer advanced Cybersecurity content — suggesting high intent but price sensitivity. Strategy: discount offers, flash sales, and free trial content in their preferred niche.

Focused Achievers (25.5%): Moderate enrollment with exceptionally high Learning Depth Index (1.45 vs cluster avg 0.27). Systematically advance through difficulty levels. Strategy: learning path roadmaps, skill-based badges, and progressive course sequences.

5.3 Evaluation Metrics

Metric	Value	Interpretation
Silhouette Score (K-Means, k=4)	0.1142	Moderate — acceptable for high-dimensional behavioral data
Silhouette Score (Hierarchical, k=4)	~0.11	Consistent with K-Means, confirms stability
Inertia (K-Means, k=4)	21,142	Used in Elbow analysis; plateau confirms k=4
Engagement Lift — Power Learners	+58% courses, +64% spend	Highest-value segment significantly above average
Engagement Lift — Budget Browsers	-39% courses, -59% spend	Clearly distinguishable low-engagement cluster

6. Streamlit Web Application

The interactive dashboard was built using Streamlit and provides five core modules:

Module	Functionality
Overview & EDA	Gender pie, age histogram, category/level bar charts, spending distribution, enrollment heatmap
Clustering Analysis	Elbow curve, silhouette bar chart, cluster size distribution, profile table, feature comparison
Learner Profile Explorer	UserID selector, segment badge, behavioral metrics, enrolled courses, learner vs cluster avg chart
Course Recommender	Cluster-aware top-N recommendations with relevance score bars and peer enrollment counts

Segment Comparison	Side-by-side segment cards, spending overlap, category preference breakdown, engagement lift table

7. Conclusion

This project successfully introduces student-centric intelligence to the EduPro platform. By shifting focus from predicting course demand to understanding learner journeys, the system enables EduPro to deliver meaningful, adaptive, and engaging learning experiences.

The K-Means segmentation framework identifies four actionable learner profiles with distinct behavioral characteristics, enabling targeted marketing, personalized recommendations, and tailored learning path design. The cluster-aware recommendation engine — combining content-based filtering with peer popularity signals — provides a scalable, interpretable solution that avoids the cold-start problem.

Future work could incorporate temporal analysis of learning progression, deep learning-based embedding models for richer course-learner matching, and A/B testing frameworks to measure recommendation impact on completion rates and platform retention.

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